DIFFUSION OF INNOVATIONS

Fourth Edition

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innovation, through diffusion and adoption of the innovation by users, to its consequences. Recognition of a problem or need may happen by means of a political process through which a social problem rises to a high priority on the agenda of problems that deserve research. In other cases, a scientist may perceive a future problem or sense a present difficulty and begin a research program to seek solutions.

Many, but not all, technical innovations come out of research. *Basic research* is defined as original investigations for the advancement of scientific knowledge that do not have the specific objective of applying this knowledge to practical problems. The results of basic research are used in *applied research*, which consists of scientific investigations that are intended to solve practical problems.

The usual next phase in the innovation-development process is *development*, defined as the process of putting a new idea into a form that is expected to meet the needs of an audience of potential adopters. The next phase, *commercialization*, is defined as the production, manufacturing, packaging, marketing, and distribution of a product that embodies an innovation. Commercialization is usually done by private firms, as the name of this phase implies.

A particularly crucial point in the innovation-development process is the decision to begin diffusing an innovation to potential adopters. This choice represents an arena in which researchers come together with change agents. How are innovations evaluated? One way is through *clinical trials*, scientific experiments that are designed to determine prospectively the effects of an innovation in terms of its efficacy, safety, and the like.

Finally, the innovation diffuses, is adopted, and eventually causes consequences, the final step in the innovation-development process. The six phases described here may not always occur in a linear sequence, the time-order of the phases may be different, or certain phases may not occur at all.

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THE INNOVATION-DECISION PROCESS

> One must learn by doing the thing, for though you think you know it, you have no certainty until you try.
>
> —Sophocles, 400 B.C.

The *innovation-decision process* is the process through which an individual (or other decision-making unit) passes (1) from first knowledge of an innovation, (2) to forming an attitude toward the innovation, (3) to a decision to adopt or reject, (4) to implementation of the new idea, and (5) to confirmation of this decision. This process consists of a series of actions and choices over time through which an individual (or an organization) evaluates a new idea and decides whether or not to incorporate the innovation into ongoing practice. This behavior consists essentially of dealing with the uncertainty that is inherently involved in deciding about a new alternative to those previously in existence. The perceived newness of an innovation, and the uncertainty associated with this newness, is a distinctive aspect of innovation decision making, compared to other types of decision making.

This chapter describes a model of the innovation-decision process and five stages in this process, and summarizes the research evidence that these stages exist. Our main concern here is with optional innovation-decisions that are made by individuals, although much of what is said contributes a basis for our later discussion of the innovation-decision process in organizations (Chapter 10).

A Model of the Innovation-Decision Process

Diffusion scholars have long recognized that an individual's decision about an innovation is not an instantaneous act. Rather, it is a *process* that occurs over time, consisting of a series of actions and decisions. What is the nature of these sequential stages in the process of innovation decision making?

Our present model of the innovation-decision process is depicted in Figure 5–1. This conceptualization consists of five stages:

1. *Knowledge* occurs when an individual (or other decision-making unit) is exposed to an innovation's existence and gains some understanding of how it functions.
2. *Persuasion* occurs when an individual (or some other decision-making unit) forms a favorable or unfavorable attitude toward the innovation.
3. *Decision* occurs when an individual (or some other decision-making unit) engages in activities that lead to a choice to adopt or reject the innovation.
4. *Implementation* occurs when an individual (or other decision-making unit) puts an innovation into use.
5. *Confirmation* occurs when an individual (or some other decision-making unit) seeks reinforcement of an innovation-decision already made, or reverses a previous decision to adopt or reject the innovation if exposed to conflicting messages about the innovation.

Now we describe the behaviors that occur at each of the five stages in the innovation-decision process.

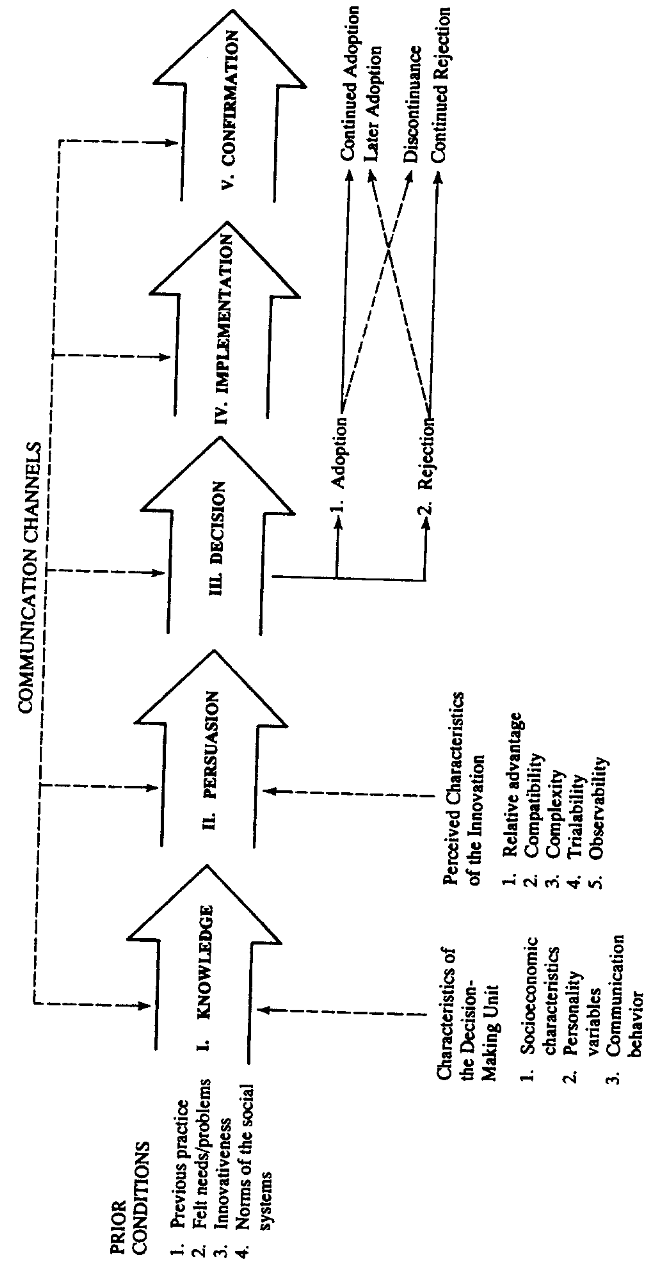
Knowledge Stage

The innovation-decision process begins with the *knowledge* stage which occurs when an individual (or other decision-making unit) is exposed to an innovation's existence and gains some understanding of how it functions.

Which Comes First, Needs or Awareness of an Innovation?

Some observers claim that an individual plays a passive role in being exposed to awareness-knowledge about an innovation. It is argued that an individual becomes aware of an innovation by accident, since the individual cannot actively seek an innovation until he/she knows that it exists. For example, Coleman and others (1966) concluded that initial knowledge about a new medical drug mainly occurred through communication channels and messages (such as salespersons and advertising) that physi-

Figure 5–1. A Model of Stages in the Innovation-Decision Process



The *innovation-decision process* is the process through which an individual (or other decision-making unit) passes from first knowledge of an innovation, to forming an attitude toward the innovation, to a decision to adopt or reject, to implementation of the new idea, and to confirmation of this decision.

cians did not seek; at later stages in the innovation-decision process, however, doctors became active information-seekers, usually from their peers in communication networks.

Other scholars of diffusion feel that an individual gains awareness-knowledge through behavior that must be initiated, and that awareness-knowledge is not just a passive activity. The predispositions of individuals influence their behavior toward communication messages and the effects that such messages are likely to have. Individuals tend to expose themselves to ideas that are in accordance with their interests, needs, and existing attitudes. Individuals consciously or unconsciously avoid messages that are in conflict with their predispositions. This tendency is called selective exposure.* Hassinger (1959) argued that individuals will seldom expose themselves to messages about an innovation unless they first feel a need for the innovation, and that even if such individuals are exposed to these innovation messages, such exposure will have little effect unless the innovation is perceived as relevant to the individual's needs and as consistent with the individual's attitudes and beliefs.† For example, a farmer can drive past a hundred miles of hybrid corn in Iowa and never "see" the innovation. A Californian can walk past a house with solar panels on the roof and not perceive this innovation. Selective exposure and selective perception act as particularly tight shutters on the windows of our minds in the case of innovation messages, because such ideas are new. We cannot have consistent and favorable attitudes about ideas that we have not previously encountered. There is, then, much in the ideas of selective exposure and selective perception to support Hassinger's viewpoint that the need for an innovation must usually precede awareness-knowledge of the innovation.

How are needs created? A *need* is a state of dissatisfaction or frustration that occurs when one's desires outweighs one's actualities, when "wants" outrun "gets." An individual may develop a need when he or she learns that an innovation exists. Therefore, innovations *can* lead to needs, as well as vice versa. Some change agents create needs among their clients through pointing out the existence of desirable new ideas. Thus knowledge of the existence of an innovation can create motivation for its adoption.

**Selective exposure* is the tendency to attend to communication messages that are consistent with one's existing attitudes and beliefs.

†This is *selective perception*, the tendency to interpret communication messages in terms of one's existing attitudes and beliefs.

By no means are perceived needs or problems a very complete explanation of why individuals begin the innovation-decision process. In part, this is because individuals do not always recognize when they have a problem, nor do individuals' needs always agree with what experts might think the individuals need. Professor Edgar Dale was fond of saying: "We may want food and not need it. And we may need vitamins and minerals and fail to want them."

Does a need precede knowledge of a new idea, or does knowledge of an innovation create a need for a new idea? Perhaps this is a chicken-or-egg problem. Research does not provide a clear answer to this question of whether awareness of a need or awareness of an innovation (that creates a need) comes first. The need for certain innovations, such as a pesticide to treat a new bug that is destroying a farmer's crops, probably comes first. But for many other new ideas, the innovation may create the need. This sequence may be especially likely for consumer innovations like clothing fashions and electronic products like compact discs.

Types of Knowledge About an Innovation

The innovation-decision process is essentially an information-seeking and information-processing activity in which the individual is motivated to reduce uncertainty about the advantages and disadvantages of an innovation. The individual wishes to understand the innovation, and to give meaning to it. This is a social process, involving talking with others.

An innovation typically contains *software information*, which is embodied in the innovation and which serves to reduce uncertainty about the cause-effect relationships that are involved in achieving a desired outcome (such as meeting a need of the individual). Questions such as "What is the innovation?" "How does it work?" and "Why does it work?" are the main concerns of an individual about an innovation. The first of these three types of knowledge, *awareness-knowledge*, is information that an innovation exists. Awareness-knowledge then motivates an individual to seek "how-to" knowledge and principles knowledge. This type of information-seeking is concentrated at the knowledge stage of the innovation-decision process, but it may also occur at the persuasion and decision stages.

How-to knowledge consists of information necessary to use an innovation properly. The adopter must understand what quantity of an innovation to secure, how to use it correctly, and so on. In the case of innovations that are relatively more complex, the amount of how-to knowledge needed for proper adoption is much greater than in the case of less complex ideas. And when an adequate level of how-to knowledge is not obtained prior to the trial and adoption of an innovation, rejection and discontinuance are likely to result. To date, few diffusion investigations are available that deal with how-to knowledge.

Principles-knowledge consists of information dealing with the functioning principles underlying how the innovation works. Examples of principles-knowledge are: The notion of germ theory, which underlies the functioning of water boiling, vaccinations, and latrines in village sanitation and health campaigns; the fundamentals of human reproduction, which form a basis for family planning innovations; and the biology of plant growth, which underlies the adoption of fertilizer by farmers and gardeners. It is usually possible to adopt an innovation without principles-knowledge, but the danger of misusing the new ideas is greater, and discontinuance may result. Certainly, the competence of individuals to decide whether or not to adopt an innovation is facilitated by principles know-how. If a problem occurs in an individual's use of an innovation, principles-knowledge may be essential in solving it.

What is the role of change agents in bringing about the three types of knowledge? Most change agents concentrate their efforts on creating awareness-knowledge, although this goal often can be achieved more efficiently in many client systems by mass media channels. Change agents could perhaps play their most distinctive and important role in the innovation-decision process if they concentrated on how-to knowledge, which is probably most essential to clients in their trial of an innovation (at the decision stage in the innovation-decision process). Most change agents perceive that creation of principles-knowledge is outside the purview of their responsibilities and is a more appropriate task for formal schooling. It is often too complex a task for change agents to teach basic understanding of principles. But when such understanding is lacking, the change agent's long-run task is often more difficult.

Early Versus Late Knowers of Innovations

The following generalizations summarize the results of findings regarding early knowing about an innovation:

5–1. *Earlier knowers of an innovation have more formal education than later knowers.*

5–2. *Earlier knowers of an innovation have higher socioeconomic status than late knowers.*

5–3. *Earlier knowers of an innovation have more exposure to mass media channels of communication than later knowers.*

5–4. *Earlier knowers of an innovation have more exposure to interpersonal channels than later knowers.*

5–5. *Earlier knowers of an innovation have more change agent contact than later knowers.*

5–6. *Earlier knowers of an innovation have more social participation than later knowers.*

5–7. *Earlier knowers of an innovation are more cosmopolite than later knowers.*

The characteristics of earlier knowers of an innovation are similar to the characteristics of innovators: More formal education, higher socioeconomic status, and the like. But earlier knowers are not necessarily innovators.

Knowing about an innovation is often quite different from using a new idea. Most individuals know about many innovations that they have not adopted. Why? One reason is because an individual may know about a new idea but not regard it as relevant to the individual's situation, and as potentially useful. Attitudes toward an innovation, therefore, frequently intervene between the knowledge and decision functions. In other words, the individual's attitudes or beliefs about the innovation have much to say about his or her passage through the innovation-knowledge process. Consideration of a new idea does not go beyond the knowledge function if an individual does not define the information as relevant to his or her situation or if sufficient knowledge is not obtained to become adequately informed so that persuasion can take place.

Persuasion Stage

At the *persuasion** stage in the innovation-decision process, the individual (or some other decision-making unit) forms a favorable or unfavor-

*We do not define persuasion with exactly the same connotation as many other communication scholars, who use the term to imply a source's communication with an intent to induce attitude change in a desired direction on the part of a receiver. Our meaning for persuasion is equivalent to attitude formation and change on the part of an individual, but not necessarily in the direction intended by some particular source, such as a change agent.

able attitude* toward the innovation. Whereas the mental activity at the knowledge stage was mainly cognitive (or knowing), the main type of thinking at the persuasion function is affective (or feeling). Until the individual knows about a new idea, of course, he or she cannot begin to form an attitude toward it.

At the persuasion stage the individual becomes more psychologically involved with the innovation; he or she actively seeks information about the new idea, *what* messages he or she receives, and *how* he or she interprets the information that is received. Thus, selective perception is important in determining the individual's behavior at the persuasion stage, for it is at the persuasion stage that a general perception of the innovation is developed. Such perceived attributes of an innovation as its relative advantage, compatibility, and complexity are especially important at this stage (see Figure 5–1).

In developing a favorable or unfavorable attitude toward the innovation, an individual may mentally apply the new idea to his or her present or anticipated future situation before deciding whether or not to try it. This is a kind of vicarious trial. The ability to think hypothetically and counter-factually and to project into the future is an important mental capacity at the persuasion stage where forward planning regarding the innovation is involved.

All innovations carry some degree of uncertainty for the individual, who is typically unsure of the new idea's results and thus feels a need for social reinforcement of his or her attitude toward the new idea. The individual wants to know that his or her thinking is on the right track, in comparison with the opinion of peers. Mass media messages are too general to provide the specific kind of reinforcement that the individual needs to confirm the individual's beliefs about the innovation.

At the persuasion stage, and especially at the decision stage, an individual is motivated to seek *innovation-evaluation information*, the reduction in uncertainty about an innovation's expected consequences. Here an individual usually wants to know the answers to such questions as "What are the innovation's consequences?" and "What will its advantages and disadvantages be in my situation?" This type of information, while often easily available from scientific evaluations of an innovation, is usually sought by most individuals from their near-peers whose subjective opinion of the innovation (based on their personal experience with adoption of the new idea) is most convincing. When someone who is like us tells us of their positive evaluation of a new idea, we are often motivated to adopt it.

Attitude is a relatively enduring organization of an individual's beliefs about an object that predisposes his or her actions.

The main outcome of the persuasion stage in the innovation-decision process is either a favorable or unfavorable attitude toward the innovation. It is assumed that such persuasion will lead to a subsequent change in overt behavior (that is, adoption or rejection) consistent with the attitude held. But in many cases attitudes and actions are quite disparate. For example, such a discrepancy between favorable attitudes and actual adoption is frequently found for contraceptive ideas in Third World nations. For instance, surveys of parents of childbearing age in many of these nations show that almost all of these individuals say they are informed about family planning methods and have a favorable attitude toward using them. But only 15 or 20 percent of the parents have actually adopted contraceptives (Rogers, 1973). This attitude-use discrepancy is called the "KAP-gap" (KAP refers to knowledge-attitude-practice) in the family planning field. One reason this gap occurs is because the available family planning methods are not acceptable to parents, owing to certain undesirable side-effects that are associated with them in the minds of potential adopters.

In many other circumstances, formation of a favorable or unfavorable attitude toward an innovation does not lead to an adoption or rejection decision. Nevertheless, there is a tendency in this direction, that is, for attitudes and behavior to become consistent. Why aren't attitudes and actions more closely related? In some cases overt behavior change is beyond an individual's control. Let's say that an individual intends to get a blood test for HIV/AIDS tomorrow but the health clinic is closed. Adoption of a new idea often entails obtaining the innovation in the form of a product or service, which may not always be available to an individual, or which may be too expensive for the individual to adopt. Such intervening factors can cause a KAP-gap.

Another reason why attitudes are not converted into action may be the communication channels utilized by an individual who is a potential adopter. We know from past research that interpersonal communication from a near-peer who is a satisfied adopter often pushes a potential adopter over the edge of decision into adoption of an innovation. Some potential adopters are so located in the diffusion network of their system that they do not have interpersonal communication about the innovation with a near-peer. Perhaps the individual is a social isolate in the network, or else just happens not to have any satisfied adopters as near-peers. Even though this individual has favorable attitudes toward the innovation, adoption may not occur until the individual has interpersonal communication with a satisfied adopter.

Certain individuals are more likely to have an attitudes/adoption gap than are others. For example, later in this chapter we show that earlier adopters of an innovation are characterized by a relatively shorter innovation-decision period (the length of time from awareness-knowledge of an innovation to its adoption). In comparison, later adopters require a longer time period, once aware of a new idea, to move to adoption. So at any given point in time, these later adopters are likely to have a discrepancy between knowledge/attitudes versus adoption. They may have low *efficacy*, defined as the degree to which an individual feels they can control their future. Someone with relatively low self-efficacy would not possess the self-confidence to think that they could adopt the innovation.

An attitude-adoption gap is particularly characteristic of certain innovations, such as new ideas that are preventive in nature. A *preventive innovation* is a new idea that an individual adopts in order to avoid the possible occurrence of some unwanted event in the future (see Chapter 6). The undesired event may, or may not, occur if the innovation is not adopted. So the desired consequences of a preventive innovation are relatively uncertain. Further, adoption and use of a preventive innovation may be unpleasant (such as getting a blood test for HIV/AIDS). Under such circumstances, the individual's motivation to adopt the innovation is weak. Examples of preventive innovations are contraceptives, getting a mammogram test for breast cancer, stopping smoking, the use of automobile seat belts, buying insurance, and preparing a preventive for a possible disaster such as an earthquake or hurricane. Even when an individual perceives a need for innovation, and when the innovation is accessible, adoption often does not occur. So the rate of adoption of preventive innovations is relatively slower than for nonpreventive innovations.

The persuasion-adoption discrepancy for preventive innovations can sometimes be closed by a *cue-to-action*, an event occurring at a certain time that crystallizes a favorable attitude into overt behavior change. Some cues-to-action occur naturally; for instance, many women adopt a contraceptive after they experience a pregnancy scare or an abortion (Rogers, 1973). In other cases, a cues-to-action can be created by a change agency; for instance, some national family planning programs pay incentives in order to provide a cue-to-action to potential adopters. Holding a campaign may provide a cue-to-action for certain individuals. An example is the one-day national "Smoke Out," held each November in the United States, in which thousands of people stop smoking.

Decision Stage

The *decision* stage in the innovation-decision process occurs when an individual (or other decision-making unit) engages in activities that lead to a choice to adopt or reject an innovation. *Adoption* is a decision to make full use of an innovation as the best course of action available. *Rejection* is a decision not to adopt an innovation.

One way to cope with the inherent uncertainty about an innovation's consequences is to try out the new idea on a partial basis. Most individuals will not adopt an innovation without trying it first on a probationary basis in order to determine its usefulness in their own situation. This small-scale trial is often part of the decision to adopt. In some cases, an innovation cannot be divided for trial and so it must be adopted or rejected in toto. Innovations that can be divided for trial are generally adopted more rapidly (see Chapter 6). Most individuals who try an innovation then move to an adoption decision, if the innovation has at least a certain degree of relative advantage. Methods to facilitate the trial of innovations such as the distribution to clients of free samples of a new idea, will speed up the rate of adoption. A field experiment among Iowa farmers found that the free trial of a new weed spray speeded the innovation-decision period by about a year (Klonglan, 1962, 1963; Klonglan and others, 1960a).

For some individuals and for some innovations, the trial of a new idea by a peer like themselves can substitute, at least in part, for their own trial of an innovation. This "trial-by-others" provides a kind of vicarious trial for an individual. Change agents often seek to speed up the innovation-process for individuals by sponsoring demonstrations of a new idea in a social system (see Chapter 9). This demonstration can be quite effective in influencing adoption by individuals, especially if the demonstrator is an opinion leader (Magill and Rogers, 1981).

The innovation-decision process can just as logically lead to a rejection decision as to adoption. In fact, each stage in the innovation-decision process is a potential rejection point. For instance, it is possible to reject an innovation at the knowledge stage by simply forgetting about it after gaining initial awareness-knowledge. And, of course, rejection can occur even after a prior decision to adopt. This is discontinuance, which usually occurs in the confirmation stage of the innovation-decision process. Two different types of rejection are (Eveland, 1979):

1. *Active rejection*, which consists of considering adoption of the innovation (including even its trial) but then deciding not to adopt it.
2. *Passive rejection* (also called nonadoption), which consists of never really considering the use of the innovation.

Obviously, these two types of rejection represent quite different types of behavior. Unfortunately, they have not been distinguished in past diffusion researches. Perhaps owing to the pro-innovation bias that pervades much diffusion inquiry (Chapter 3), investigation of rejection behavior of all kinds has not received much scientific attention.

Further, there is usually an implicit assumption in diffusion studies of a linear sequence of the first three stages in the innovation-decision process: Knowledge-persuasion-decision. In some cases, the actual sequence of stages may be knowledge-decision-persuasion. For example, in a Korean village that I once studied, a meeting of married women was called, and, after a lecture by a government family planning official about the IUD (intrauterine device), there was a show of hands to indicate the women who wanted to adopt (Rogers and Kincaid). Eighteen women (of the forty present) volunteered, and promptly marched off to a nearby health clinic to have IUDs inserted. In this case, a presumably optional innovation-decision almost became a collective innovation-decision as a result of strong group pressure. A similar group-oriented strategy for family planning is followed in the "group planning of births" in the People's Republic of China (Rogers and Chen, 1980). The community decides who should have babies in a group meeting held annually, and then parents are influenced to follow these group birth plans. Such strong group pressure for adoption of an innovation would be abhorrent to values on individual freedom in many cultures, but they are not in Korea and China. So the knowledge-persuasion-decision sequence proposed in our model of the innovation-decision process may be somewhat culture-bound. In some sociocultural settings, the knowledge-decision-persuasion sequence may frequently occur, at least for certain innovations.

Implementation Stage

Implementation occurs when an individual (or other decision-making unit) puts an innovation into use. Until the implementation stage, the innovation-decision process has been a strictly mental exercise. But implementation involves overt behavior change, as the new idea is actually put into practice. It is often one thing for an individual to decide to adopt a new idea, but quite a different thing to put the innovation into use. Problems in exactly how to use the innovation crop up at the implementation stage. Implementation usually follows the decision stage rather directly unless it is held up by some logistical problem, like the temporary unavailability of the innovation.

A certain degree of uncertainty about the expected consequences of the innovation still exists for the individual at the implementation stage, even though the decision to adopt has been made previously. When it comes to implementation, an individual particularly wants to know the answers to such questions as "Where do I obtain the innovation?" "How do I use it?" "How does it work? " and "What operational problems am I likely to encounter, and how can I solve them?" So much active information-seeking usually takes place at the implementation stage. Here the role of the change agent is mainly to provide technical assistance to the client as he or she begins to use the innovation.

Problems of implementation are much more serious when the adopter is an organization rather than an individual. In an organizational setting, a number of individuals are usually involved in the innovation-decision process, and the implementers are often a different set of people from the decision makers. Also, the organizational structure that gives stability and continuity to an organization may be a resistant force to implementation of an innovation. As we show in Chapter 10, it was not until diffusion scholars began to study the innovation-decision process in organizations that the importance of the implementation stage for individual/optional innovation-decisions became apparent. The implementation of innovations in organization has been heavily studied in the 1980s and early 1990s.

When does the implementation stage end? It may continue for a lengthy period of time, depending on the nature of the innovation. But eventually a point is reached at which the new idea becomes an institutionalized and regularized part of the adopter's ongoing operations. The innovation finally loses its distinctive quality as the separate identity of the new idea disappears. This point is usually considered the end of the implementation stage, and is often referred to as routinization or institutionalization, especially when it occurs in organizations (see Chapter 10).

Implementation may also represent the termination of the innovation-decision process, at least for most individuals. But for others, a fifth stage of confirmation may occur, as we explain in a following section. First, we shall discuss the concept of re-invention, which often occurs at the implementation stage.

Re-Invention

In the early years of diffusion study, we assumed that adoption of an innovation meant the exact copying or imitation of how the innovation had been used previously in a different setting. Sometimes the adoption of an innovation does indeed represent identical behavior; for example, the California Fair Trade Law of 1931, the first law of its kind, was adopted by ten other states complete with three serious typographical errors that had appeared in the California bill (Walker, 1971). In many other cases, however, an innovation is not invariant as it diffuses. The new idea changes and evolves during the diffusion process.

Diffusion scholars now recognize the concept of *re-invention*, defined as the degree to which an innovation is changed or modified by a user in the process of its adoption and implementation. Until the 1970s, re-invention was ignored completely, or was considered at most a very infrequent behavior. When a respondent in a diffusion survey told about his or her re-invention of a new idea, it was considered as a very unusual kind of behavior, and was treated as a "noise" in diffusion research. Adopters were considered to be passive acceptors of an innovation, rather than active modifiers and adapters of a new idea. Once diffusion scholars made the mental breakthrough of recognizing that re-invention could happen, they began to find that quite a lot of it occurred.

Re-invention could not really be investigated until diffusion researchers began to gather data about implementation, for most re-invention occurs at the implementation stage of the innovation-decision process. We now know that a great deal of re-invention occurs for certain innovations which suggests that previous diffusion research may have erred, by measuring adoption as a stated intention to adopt (at the decision stage). The fact that re-invention often happens is a strong argument for measuring adoption at the implementation stage, and as action by the adopter, rather than just as an intention to act.

Most scholars in the past have made a distinction between invention and innovation. *Invention* is the process by which a new idea is discovered or created, while adoption is a decision to make full use of an innovation. Thus adoption of an innovation is the process of *using* an existing idea, which may have been previously invented by someone. This heuristic difference between invention and innovation, however, is not so clear-cut once we acknowledge that an innovation is not necessarily a fixed entity as it diffuses within a social system. "Re-invention" seems like an appropriate word to describe the degree to which an innovation is changed or modified by the user in the process of its adoption and implementation.

How Much Re-Invention Occurs?

The recent research focus on re-invention was launched by Charters and Pellegrin (1972), who were the first scholars to recognize the occurrence of re-invention (although they did not use the term per se). These researchers traced the adoption and implementation of the educational innovation of "differentiated staffing" in four schools over a one-year period. They concluded that "differentiated staffing was little more than a word for most [teachers and administrators], lacking concrete parameters with respect to the role performance of participants. . . . The word could (and did) mean widely differing things to the staff, and nothing to some. . . . The innovation was to be invented on the inside, not implemented from the outside." These scholars noted the degree to which the innovation was shaped differently in each of the four organizations they studied.*

When an invention is designed with the concept of re-invention in mind, a certain degree of re-invention often occurs as the innovation diffuses. For instance, previous research on innovation in organizations had assumed that a new technological idea enters a system from external sources and is then adopted (with relatively little adaptation of the innovation) and implemented as part of the organization's ongoing operations. Thus it is assumed that adoption of an innovation by individual A or organization A will look much like adoption of this same innovation by individual B or organization B. Recent investigations call this assumption into serious question. For instance:

1. A national survey of schools adopting educational innovations promoted by the National Diffusion Network, a decentralized diffusion system, found that 56 percent of the adopters implemented only selected aspects of an innovation. Much such re-invention was relatively minor, but in 20 percent of the adoptions, important changes were made in the innovations (Emrick and others, 1977).

*Charters and Pellegrin became aware of re-invention (1) because they used a process research approach, and (2) due to their focus on the implementation stage in the innovation-decision process for the innovation of differentiated staffing.

2. An investigation of 111 innovations in scientific instruments field by Von Hippel (1976) found that in about 80 percent of the cases, the innovation process was dominated by the user (that is, a customer). The user typically built a prototype model of the new product and then turned it over to a manufacturer for volume production. So the "adopters" played a very important role in designing and redesigning these industrial innovations.
3. Of the 104 adoptions of innovations by mental health agencies that were studied in California, re-invention occurred somewhat more often (in fifty-five cases) than did unchanged adoption (in forty-nine cases) (Larsen and Agarwala-Rogers, 1977, p. 37; 1977b).
4. A study of the adoption by fifty-three local government agencies of a computer-based planning tool (called GBF/DIME) that was promoted to them by a Federal agency, found that about half of the "adoptions" represented at least some degree of re-invention (Eveland and others, 1977; Rogers and others, 1977a).
5. Research on the rapid diffusion of a school-based drug abuse prevention program called D.A.R.E. (Drug Abuse Resistance Education) disclosed a high degree of re-invention. D.A.R.E. began in 1983 in Los Angeles, and ten years later, over 5 million fifth and sixth graders were taught the seventeen-lesson D.A.R.E. curriculum in their classroom by a uniformed policeman. This speedy rate of adoption occurred because the drug problem was high on the public agenda in the late 1980s. A good deal of re-invention took place (Rogers, 1993a, pp. 139–162). For instance, one of the seventeen D.A.R.E. lessons encouraged school children not to join gangs. However, schools in which there was not a gang problem did not teach the D.A.R.E. material on that topic. The basic idea of D.A.R.E., that of having a uniformed policeman teach the drug prevention lessons, however, was used by all adopters.

Generalization 8 states that: *At least some degree of re-invention occurs at the implementation stage for many innovations and for many adopters.*

Re-Invention Is Not Necessarily Bad

Whether re-invention is considered good or bad depends on one's point of view. Re-invention generally does not receive much favorable attention from research and development agencies, who may consider re-invention a distortion of their original technology. In fact, some designers of innovation shape a new idea so that it is particularly difficult to re-invent. They feel that "re-invention proofing" is a means of maintaining

quality control over their innovation. Diffusion agencies may also be unfavorable toward re-invention, feeling that they know best as to the form of the innovation that the users should adopt. Change agents find it difficult to measure their performance if an innovation they are promoting changes over time and across different adopters. Their usual measure, the rate of adoption of an innovation, can become an ambiguous index when a high degree of re-invention occurs. In extreme cases of re-invention, the original innovation may even lose its identity.

Adopters, on the other hand, generally think that re-invention is a desirable quality. They emphasize or even overemphasize the amount of re-invention that they have accomplished (Rice and Rogers, 1980). The choices available to a potential adopter are not just adoption or rejection; modification of the innovation or selective rejection of some components of the innovation may also be options (as in the case of D.A.R.E.). Some implementation problems by an individual or an organization are unpredictable by nature, so changes in the originally planned innovation often should occur.

Re-invention often is beneficial to the adopters of an innovation. Flexibility in the process of adopting an innovation may reduce mistakes and encourage customization of the innovation to fit it more appropriately to local situations or changing conditions. As a result of re-invention, an innovation may be more appropriate in matching an adopter's preexisting problems and more responsive to new problems that arise during the innovation-decision process. A national survey of innovation in public schools found that when an educational innovation was re-invented by a school, its adoption was more likely to be continued and less likely to be discontinued (Berman and Pauley, 1975). Discontinuance happened less often because the re-invented innovations better fit a school's circumstances. This investigation disclosed that a rather high degree of re-invention occurred: The innovations and the schools engaged in a kind of mutually influencing interaction, as the new idea and the school moved closer to each other (Berman and McLaughlin, 1974, 1975, 1978; Berman and others, 1975, 1977). Usually, the school changed very little, and the innovation substantially.*

*Eveland and others (1977) used one procedure for measuring the degree of re-invention: They identified the number of elements in each implementation of an innovation that were similar to, or different from, the "mainline" version of the innovation (that was promoted by a change agency). Most innovations can be decomposed analytically into constituent elements, thus offering one means of indexing the degree of re-invention.

Why Does Re-Invention Occur?

Some of the reasons for re-invention lie in the innovation itself, while others involve the individual or organization that is adopting the new idea.

1. Innovations that are relatively more complex and difficult to understand are more likely to be re-invented (Larsen and Agarwala-Rogers, 1977). Re-invention thus may be a simplification of the innovation.

2. Re-invention can occur because of an adopter's lack of full knowledge about the innovation, such as when there is a relatively little direct contact between the adopter and the change agents or previous adopters (Rogers and others, 1977; Eveland and others, 1977; Larsen and Agarwala-Rogers, 1977, p.38). For example, re-invention of a geographically based computer software system (GBF/DIME) occurred more frequently when change agents only created awareness-knowledge of the innovation, than when consultation was provided to the adopting organization at the implementation stage. So re-invention sometimes happens due to ignorance and inadequate learning.

3. An innovation that is an abstract concept or that is a tool (like a computer software program) with many possible applications is more likely to be re-invented (Rogers, 1978). The elements comprising an innovation may be tightly or loosely bundled or packaged (Koontz, 1976). A tightly bundled innovation is a collection of highly interdependent components; it is difficult to adopt one element without adopting the other elements. A loosely bundled innovation consists of elements that are not highly interrelated; such an innovation can be flexibly suited by adopters to their conditions. So the designer or manufacturer of an innovation can affect the degree of re-invention by making the innovation easy or difficult to re-invent (Von Hippel and Finkelstein, 1979).

4. When an innovation is implemented in order to solve a wide range of users' problems, re-invention is more likely to occur. A basic reason for re-invention is that one individual or organization applies the innovation to a different problem than does another. The perceived problem that originally motivated an individual to search for an innovation determines in part how the innovation will be used. The degree of re-invention of an innovation is greater when a wide degree of heterogeneity exists in the individual and organizational problems with which the innovation is matched.

5. Local pride of ownership of an innovation may also be a cause of re-invention. Here the innovation is often modified in certain rather cosmetic or minor ways so that it appears to be a local product. In some cases of such pseudo-re-invention, the innovation may just be given a new name, without any fundamental changes in the innovation itself. Such localization may be motivated by a desire for status on the part of the adopter, or by a desire to make the innovation more acceptable to individuals in the local system. Often, "Locals say that the innovation is local," as Havelock (1974) found in a survey or 353 U.S. school superintendents. Perhaps as one observer suggested, an innovation may be somewhat like a toothbrush in that people do not like to borrow it from someone else. They want their own. Or at least they want to put their own "bells and whistles" on the basic innovation, so that it looks different from others' adoption of the innovation. A strong psychological need to re-invent seems to exist for many individuals.

An illustration is provided by the diffusion of computers to local government in the United States, who used them for such data-handling tasks as accounting, issuing payrolls, and recordkeeping. A high rate of re-invention occurred when twelve cities and counties adopted the innovation of computer data processing (Danziger 1977). Computer programmers working for a local government viewed such modifications of software packages as a challenging and creative task. It was more fun to re-invent a computer software program than simply to transfer it from another local government or to purchase it from a commercial supplier, which was viewed as unstimulating drudgery. Further, Danziger (1977) found that local government officials emphasized their degree of re-invention, stressing the uniqueness of their adoption. The relatively petty "bells and whistles" that the adapters had re-invented appeared to them to be major improvements.

6. Finally, re-invention may occur because a change agency encourages its clients to modify an innovation. While most change agencies generally oppose re-invention, decentralized diffusion systems may encourage their clients to re-invent new ideas. For example, the diffusion of D.A.R.E. was not managed or directed by a federal agency, so each school was left to do its own thing with the seventeen-lesson D.A.R.E. curriculum (Rogers, 1993a).

Recognition of the existence of re-invention brings into focus a different view of adoption behavior: Instead of simply accepting or rejecting an innovation as a fixed idea, potential adopters on many occasions are active participants in the adoption and diffusion process, struggling to give their own unique meaning to the innovation as it is applied in their local context. Adoption of an innovation is thus a process of social construction. This conception of adoption behavior, involving re-invention, is more in line with what certain respondents in diffusion research have been trying to tell researchers for many years.

Re-Invention of Horse Culture by the Plains Indians

When the European colonists encountered the Plains Indians in the vast grasslands west of the Mississippi River, the horse was the central element of Indian culture. The Indians killed buffalo from horseback, fought other tribes while mounted on their ponies, and utilized their horses to move their wigwams from place to place. But the horse was not indigenous to the Plains Indians. Instead, the horse had only been introduced to the Indian tribes by Spanish explorers around 1650. Within a decade or two, French fur traders found that men, women, and children in the Plains Indian tribes were riding horses. The Indians had copied saddles, stirrups, the crupper, and the lariat from the Spanish explorers, who, in turn had borrowed these innovations from the Moors (Arabic people from North Africa, who had previously occupied Spain for 500 years). In short, the Plains Indians "copied the whole [horse] culture from a to z" (Wissler, 1923). Extensive contact occurred with the Spanish, and the horse had originally appeared with Spanish riders, so the Plains Indians easily learned how to ride the new animal. Horse culture then spread quickly throughout the American grasslands.

But one important aspect of horse culture was re-invented: The travois. Previous to the coming of the horse, the Plains Indians used dogs as beasts of burden. Tents and baggage were transported on the travois, a kind of drag frame. So when the Indians first saw the horse, they called it a dog. Many Indian tribes in the United States still speak of the horse by a name meaning dog. They enlarged the travois, and harnessed the horse to it. Only somewhat later did the Plains Indians begin to ride on horseback.

This case illustration is based on Wissler (1914 and 1923).

Confirmation Stage

As explained previously,* a decision to adopt or reject is often not the terminal stage in the innovation-decision process. For example, Mason (1962) found that Oregon farmers sought information *after* they had decided to adopt, as well as before. At the *confirmation* stage the individual (or some other decision-making unit) seeks reinforcement of the innovation-decision already made or reverses a previous decision to adopt or reject the innovation if exposed to conflicting messages about the innovation. At the confirmation stage, the individual seeks to avoid a state of dissonance or to reduce it if it occurs.

*And as indicated by such researchers as Mason (1962) and Francis and Rogers (1960).

Dissonance

Human behavior change is motivated in part by a state of internal disequilibrium or dissonance, an uncomfortable state of mind that the individual seeks to reduce or eliminate (Festinger, 1957). An individual who feels dissonant will ordinarily be motivated to reduce this condition by changing his or her knowledge, attitudes, or actions. In the case of innovative behavior, this dissonance reduction may occur:

1. When the individual becomes aware of a felt need and seeks information about an innovation to meet this need. Here, a receiver's need for the innovation can motivate the individual's information-seeking activity about the innovation. This behavior occurs at the knowledge stage in the innovation-decision process.

2. When the individual knows about a new idea and has a favorable attitude toward it, but has not yet adopted. Then the individual is motivated to adopt the innovation by dissonance between what he or she believes versus what he or she is doing. This behavior occurs at the decision and implementation stages in the innovation-decision process.

3. After the innovation-decision to implement the innovation, when the individual secures further information that persuades him or her that he or she should *not* have adopted. This dissonance may be reduced by discontinuing the innovation. Or if he or she originally decided to reject the innovation, the individual may become exposed to pro-innovation messages, causing a state of dissonance that can be reduced by adoption. These types of behavior (discontinuance or later adoption) occur during the confirmation stage in the innovation-decision process (Figure 5–1).

These three types of dissonance reduction consist of changing behavior so that attitudes and actions are more closely in line. But it is often difficult to change one's prior decision to adopt or reject; activities have been set in motion that tend to stabilize the original decision. Perhaps a considerable expense was involved in adoption of the innovation. Therefore, individuals frequently try to avoid becoming dissonant by seeking only that information that they expect will support or confirm the decision they already made. This behavior is an example of selective exposure.* During the confirmation stage the individual wants supportive messages that will prevent dissonance from occurring. Nevertheless, some information reaches the individual that leads to questioning the adoption-rejection decision made previously in the innovation-decision process.

At the confirmation stage in the innovation-decision process, the change agent can play a special role. In the past, change agents have primarily been interested in achieving adoption decisions. At the confirmation stage they have the additional responsibility of providing supporting messages to individuals who have previously adopted. Change agents often assume that once adoption is secured, it will continue. But there is no assurance against discontinuance, because negative messages about an innovation circulate via interpersonal networks in most client systems.

Discontinuance

Discontinuance is a decision to reject an innovation after having previously adopted it. A rather surprisingly high rate of discontinuance has been found for certain innovations. Leuthold (1967) concluded from his study of a statewide sample of Wisconsin farmers that the rate of discontinuance was just as important as the rate of adoption in determining the level of adoption of an innovation at any particular time during the diffusion process. In any given year there were about as many discontinuers of an innovation as there were first-time adopters.

Two types of discontinuance are: (1) replacement and (2) disenchantment. A *replacement discontinuance* is a decision to reject an idea in order to adopt a better idea that supersedes it. In many fields, there are constant waves of innovations. Each new idea replaces an existing practice that was an innovation in its day. For example, the adoption of "gammanym" (tetracycline) led to the discontinuance of two other antibiotic drugs in the Columbia drug study (Coleman and others, 1966). Hand calculators replaced slide rules. Microcomputers replaced the use of mainframe computers. CDs (compact discs) replaced vinyl plastic for music. Many other examples of replacement discontinuances occur in everyday life.

A *disenchantment discontinuance* is a decision to reject an idea as a result of dissatisfaction with its performance. Such dissatisfaction may come

*Similarly, dissonance can be reduced by selective perception (message distortion) and by the selective forgetting of dissonant information.

about because the innovation is inappropriate for the individual and does not result in an adequate level of perceived relative advantage over alternative practice. Perhaps a government agency has ordered that the innovation is no longer safe and/or that it has side-effects that are dangerous to health. Or discontinuance may result from the misuse of an innovation that could have functioned advantageously for the individual if it had been used correctly. This later type of disenchantment seems to be more common among later adopters than among earlier adopters, who have more formal education and greater understanding of the scientific method, so they know how to generalize more carefully the results of an innovation's trial to its full-scale use. Later adopters also have fewer resources, which may either prevent adoption or cause discontinuance because the innovations do not fit their limited financial position. On the basis of past research we suggest Generalization 5–9: *Later adopters are more likely to discontinue innovations than are earlier adopters.*

Diffusion scholars previously assumed that later adopters are relatively less innovative because they did not adopt or were slower to adopt. But the evidence on discontinuance behavior suggests that many laggards adopt but then discontinue, usually owing to disenchantment. For instance, Bishop and Coughenour (1964) reported that the percentage of discontinuance for Ohio farmers ranged from 14 percent for innovators and early adopters, to 27 percent for early majority, to 34 percent for late majority, to 40 percent for laggards. Leuthold (1965) reported comparable figures of 18 percent, 24 percent, 26 percent, and 37 percent, respectively, for a sample of Canadian farmers.

High discontinuers are characterized by less formal education, lower socioeconomic status, less change agent contact, and the like, which are the opposite of the characteristics of innovators (Chapter 7). Discontinuers share the same characteristics as laggards, who indeed are characterized by a higher rate of discontinuance.

The discontinuance of an innovation is one indication that the new idea may not have been fully institutionalized and routinized into the ongoing practice and way of life of the adopter at the implementation stage of the innovation-decision process. Such routinization is less likely (and discontinuance more frequent) when the innovation is less compatible with the individual's beliefs and past experiences. Perhaps (1) there are innovation-to-innovation differences in rates of discontinuance, just as there are such differences in rates of adoption, and (2) the perceived attributes of innovations (for example, relative advantage and compatibility) are negatively related to the rate of discontinuance. For instance, we would expect an innovation with a low relative advantage to have a slow rate of adoption and a fast rate of discontinuance.

In some special cases, adopting an innovation consists mainly of discontinuing a previously adopted idea. An illustration is smoking-cessation. A lifestyle revolution has occurred in the United States in the past several decades as half of all individuals who ever smoked regularly (and who are still alive today) have stopped smoking. As a result of smoking cessation, these millions of Americans will extend their life expectancy for an average of twelve years. Smoking cigarettes is an addictive behavior, one that is extremely difficult to change in that the decision not to smoke is not fully controlled by the individual. How did this U.S. smoking revolution take place? First, such widespread discontinuance did not happen as the result of economic factors. Unlike some other nations, the U.S. government has not slapped on a high cigarette tax, which research evidence indicates can discourage young people from starting to smoke and cause some adults to stop. Canada in the early 1990s raised cigarette taxes so high that a pack cost six dollars. Some states in the U.S. have instituted higher cigarette taxes; for example, California in 1989 voted a special twenty-five cents per pack tax that is used to provide cigarette-cessation advertising and educational programs.

The main thrust for cigarette-cessation in America began with the 1964 Surgeon General's report that smoking was dangerous to health. This report was based on scientific studies of the health effects of smoking cigarettes. Warnings were placed on cigarette packs and in all cigarette advertising. In 1971, tobacco companies were stopped from running their radio and television advertising. Scientific studies of "second-hand smoke" found that nonsmokers inhaled the smoke of others, and their health suffered. City governments and private companies began to restrict places in which it was acceptable to smoke. For example, until the mid-1960s, airline flight attendants distributed free packets of cigarettes with after-meal coffee. In the late 1980s, U.S. airlines banned smoking from all domestic flights. Many restaurants provided non-smoking eating areas, and most workplaces banned smoking. Within a decade or so, the social norm on cigarette-smoking changed from one of perceiving it as a "cool" act (remember Humphrey Bogart with his ever-present cigarette dangling?) to a very negative behavior ("Only losers smoke").

Thus the lifestyle revolution of smoking-cessation in the United States was caused by advances in research-based knowledge about health risks, which provided a basis for regulations on the marketing of cigarettes and

for restrictions on where an individual can smoke (Rabin and Sugarman, 1993). Decades were necessary to change the social norms on smoking. So the widespread discontinuance of cigarette smoking in America did not happen overnight.

Forced Discontinuance and the Rise of Organic Farming

A unique and theoretically interesting type of discontinuance occurred in recent decades when federal regulatory agencies, especially the Food and Drug Administration, banned the use of certain chemical innovations. Such forced discontinuance often results from research-based evidence that a chemical innovation may cause cancer or involve some other threat to consumer health.

My Ph.D. dissertation study in 1954 was based on data gathered from 155 farmers in an Iowa farm community about their adoption of such agricultural innovations as 2, 4-D weed spray, antibiotic swine feeding supplements, and chemical fertilizers. These chemical innovations represented the wave of post–World War II agricultural technologies that were recommended to farmers by agricultural scientists at Iowa State University and by the Iowa Extension Service. Like most other diffusion investigators, I accepted the recommendations of the agricultural scientists about the chemical innovations as valid. So did most of the Iowa farmers that I interviewed in my diffusion study. I remember, however, one farmer who rejected all of these agricultural chemicals because, he claimed, they killed the earthworms and songbirds in his fields. At the time, I personally regarded his organic attitude as irrational. Certainly his farming behavior was measured as laggardly by my innovativeness scale (composed of a dozen or so agricultural innovations recommended by agricultural experts).

But the rise of the environmental movement in the United States in the 1960s and research on the long-term effects of agricultural chemicals made me begin to wonder. In 1972, the U.S. Environmental Protection Agency banned the use of DDT as an insecticide because of its threats to human health. Then DES (diethyl stilbestrol) was banned for cattle feeding, as were antibiotic swine-feeding supplements and 2, 4, 5-D weed spray. It had been found that the concentration of such chemicals increased to dangerous levels owing to biomagnification in the food chain, until levels dangerous to human health sometimes occurred.

Since the early 1980s an increasing proportion of U.S. consumers have patronized health food stores where they paid a premium price for organically grown foods. Correspondingly, the number of organic farmers and gardeners increased, as a result of growing concern about the effects of chemical pesticides and fertilizers. Organic farmers harvest somewhat lower crop yields than do chemical farmers (due to crop destruction by insects and other pests), but their costs of production are also lower and they secure a higher price for their organic food production from natural food stores.

In 1980, the U.S. Department of Agriculture reversed its policy of opposing organic farming and gardening, and began to advise U.S. farmers and gardeners to consider using fewer chemicals. The USDA also began a research program to develop appropriate seed varieties for organic farming and gardening. Surveys of organic farmers indicated that most were not "hippies," nor were they lower-educated traditional farmers. In fact, most organic farmers are commercial operators with the general characteristic of progressive farmers such as above-average education, larger farms, and so on. These characteristics are what one would expect from the innovators in adopting organic farming.

The USDA in the 1980s realized that chemical pesticides were being overused by many farmers, and launched a program called "integrated pest management" (IPM). A key factor in initiating the IPM program was the fact that hundreds of insect varieties developed resistance to existing pesticides, along with a concern for the consumer health problems resulting from biomagnification through the food chain. Integrated pest management consists of careful scouting of a farmer's fields, usually by trained scouts, who advise the farmer when a pest problem has increased above an economic threshold, and when spraying with a chemical pesticide would thus be justified. Farmers who adopt IPM typically report important savings from decreased use of pesticides. Some farmers save thousands of dollars.

Today, looking back to my 1954 Iowa diffusion investigation, the organic farmer that I interviewed certainly has had the last laugh over agricultural experts. My research procedures classified him as a laggard in 1954; by present day standards he was a superinnovator of organic farming. So for two different (and opposing) innovations, this farmer was in two extremely different adopter categories.

This case illustration is based on the author's experience.

Are There Stages in the Process?

What empirical evidence is available that the stages posited in our model of the innovation-decision process (see Figure 5–1) exist in reality? A definitive answer is difficult to provide. It is not easy for a researcher to probe the intrapersonal mental process of an individual respondent. Nevertheless, there is tentative evidence from various studies supporting the concept of stages in the innovation-decision process.

Evidence of Stages

Empirical evidence of the validity of stages in the innovation-decision process comes from an Iowa study (Beal and Rogers, 1960) that shows that most farmer-respondents recognized that they went through a series of stages as they moved from awareness-knowledge to a decision to adopt.* Specifically, they realized that they had received information about an innovation from different channels and sources at different stages in the process. Of course, it is possible for an individual to use the same sources or channels at each stage, but if he or she does not, this indicates some differentiation of the stages. Beal and Rogers (1960) also found that none of their 148 respondents reported adopting immediately after becoming aware of a new weed spray and 63 percent of the adopters of a new livestock feed reported different years for knowledge and for the decision to adopt. Most Iowa farmers seemed to require a period of time that could be measured in years to pass through the innovation-decision process. This finding provided evidence that adoption behavior is a process that contains various stages and that these stages occur over time.

Yet another type of evidence provided by Beal and Rogers (1960) deals with skipped stages. If most respondents report not having passed through a stage in the innovation-decision process for a given innovation, a question would thus be raised as to whether that stage should be included in the model. Beal and Rogers found, however, that most farmers described their behavior at each of the first three stages in the process: knowledge, persuasion, and decision. None reported skipping the knowledge or decision stages, but a few farmers did not report a trial prior to adoption.

*The conception of stages in the innovation-decision process does not necessarily require that individuals passing through the process would be conscious of what stage they are at.

How do we know that our innovation-decision process model also describes the behavior of other types of individuals than farmers as they adopt nonagriculture innovations? Coleman and others (1966) found that most physicians reported different communication channels about a new drug at the knowledge stage from those reported at the persuasion stage. LaMar (1966) studied the innovation-decision processes of 262 teachers in twenty California schools. The teachers went through the stages in the process, much as had been found in the studies of farmers. Kohl (1966) found that all fifty-eight Oregon school superintendents in his sample reported that they passed through each of the stages for such innovations as team teaching, language laboratories, and flexible scheduling.

In summary, we suggest Generalization 5–10: *Stages exist in the innovation-decision process*. The evidence is most clear-cut for the knowledge and decision stages and somewhat less so for the persuasion stage. Only rather poor data are available on the distinctiveness of the implementation and confirmation stages. Given the importance of the stages concept in diffusion research, why has more research not been directed toward understanding the innovation-decision process? Perhaps it is because the "process" nature of this research topic does not fit the "variance" type of research methods (utilizing variables measured quantitatively) used by most diffusion researchers.

Variance and Process Research

Research designed to answer the question whether stages exist in the innovation-decision process obviously needs to be quite different from the study of independent variables associated with the dependent variable of innovativeness. The first is *process research*, defined as a type of data gathering and analysis that seeks to determine the time-ordered sequence of a set of events. In contrast, *variance research* is a type of data gathering and analysis that consists of determining the co-variances among a set of variables, but not their time-order.

Most diffusion research (and most social science research) is variance-type investigation. It uses highly structured data gathering and quantitative data analysis of cross-sectional data, such as from one-shot surveys. Because only one point in time is represented in cross-sectional data, variance in a dependent variable is related to the variance in a set of independent variables. For example, variance research is appropriate for investigating variables related to innovativeness. But it cannot probe

backward in time to understand what happened first, next, and so on, and how each of these events influenced the next.

In order to explore the nature of the innovation-decision process, one needs a dynamic perspective to explain the causes and sequences of a series of events over time. Data-gathering methods for process research are less structured and the data are typically more qualitative in nature. Seldom are statistical methods used to analyze the data in process research. Diffusion scholars have frequently failed to recognize the important distinction between variance and process research in the past (Mohr, 1978 and 1982).

The general point here is that research on a topic like the innovation-decision process must be quite different from the variance research that has predominated in the diffusion field in the past.

The Hierarchy of Effects

The notion of a hierarchy of communication effects is that an individual usually must pass from knowledge change to overt behavior change in a cumulative sequence of stages that are generally parallel to the stages in the innovation-decision process (Table 5–1). Communication effects thus occur in an ordered sequence for most individuals who pass through the process. Interpersonal communication channels are generally more effective in causing the persuasion effects, and mass media channels are more effective in bringing about the knowledge effects.

A different but closely related model of stages in the innovation-decision process has been proposed by Professor James O. Prochaska, a cancer prevention researcher at the University of Rhode Island. Prochaska's five-stage model of how individuals change an addictive behavior is widely utilized in the public health field to explain the adoption of preventive health innovations like safe sex, contraception, getting a mammogram for early detection of breast cancer, and so forth. Prochaska, DiClemente, and Norcross (1992) define the five stages in changing addictive behavior as:

1. *Precontemplation*, where an individual is aware that a problem exists and begins to think about overcoming it.
2. *Contemplation*, when an individual is aware that a problem exists and is seriously thinking about overcoming it, but has not yet made a commitment to take action.

Table 5–1. The Hierarchy of Effects Corresponds to Stages in the Innovation-Decision Process and to Stages of Change

Stages in the Innovation-Decision Process	*Hierarchy of Effects*	*Porchaska's Stages of Change*
I. *Knowledge Stage*		I. Precontemplation
	1. Recall of information.	
	2. Comprehension of messages.	
	3. Knowledge or skill for effective adoption of the innovation.	
II. *Persuasion Stage*		II. Contemplation
	4. Liking the innovation.	
	5. Discussion of the new behavior with others.	
	6. Acceptance of the message about the innovation.	
	7. Formation of a positive image of the message and the innovation.	
	8. Support for the innovative behavior from the system.	
III. *Decision Stage*		III. Preparation
	9. Intention to seek additional information about the innovation.	
	10. Intention to try the innovation.	
IV. *Implementation Stage*		IV. Action
	11. Acquisition of additional information about the innovation.	
	12. Use of the innovation on a regular basis.	
	13. Continued use of the innovation.	
V. *Confirmation Stage*		V. Maintenance
	14. Recognition of the benefits of using the innovation.	
	15. Integration of the innovation into one's ongoing routine.	
	16. Promotion of the innovation to others.	

Source: Based on McGuire's (1989, p. 45) hierarchy of effects, as modified by Population Communication Services at Johns Hopkins University, and with the addition of Prochaska's stages of change (Prochaska, DiClemente, and Norcross, 1992).

3. *Preparation*, the stage at which an individual intends to take action in the immediate future, but has not yet done so.
4. *Action*, when an individual changes behavior or the environment in order to overcome the problem.
5. *Maintenance*, the stage at which an individual consolidates and continues the behavior change that was made previously.

Prochaska finds that the vast majority of addicted individuals are not at the action stage. For example he estimates that 10 to 15 percent of smokers are prepared for action, 30 to 40 percent are in the contemplation stage, and 50 to 60 percent are in the precontemplation stage. A professional change agent will be disappointed if he/she assumes all three categories are equally ready to break their smoking addiction. This misperception may be one reason why so many individuals drop out of training classes in smoking cessation, drug abuse treatment, and weight control, or else relapse to their addiction after completing the training. The more general point is that the stages in the innovation-decision process matter. An individual must pass through them in sequence.

We turn now to the role of different communication channels in the innovation-decision process.

Communication Channels by Stages in the Innovation-Decision Process

The five stages in the innovation-decision process help us understand the role of different communication channels, as is illustrated by the diffusion of gammanym among medical doctors.

Communication Channels in the Innovation-Decision Process for Tetracycline

The role of different communication channels at different stages of the innovation-decision process is provided by the classic study of the diffusion of gammanym, a new antibiotic "wonder drug" among the doctors in a medical community (Coleman and others, 1966). This innovation had spectacular results, and it was adopted very rapidly. Within two months of its release, 15 percent of the physicians had tried it; this figure reached 50 percent four months later, and at the end of seventeen months, gammanym dominated doctors' antibiotic prescriptions. Gammanym had such a striking relative advantage over previous antibiotic drugs that most of a doctor's peer networks conveyed very positive messages about the innovation. One of the most important contributions of the drug study was to establish the importance of interpersonal networks as a communication channel in the innovation-decision process.

Information that creates awareness-knowledge of an innovation seldom comes to individuals from a source or channel of communication that they actively seek. Information about a new idea can only be actively sought by individuals (1) after they are aware that the new idea exists, and (2) when they know which sources or channels can provide information about the innovation. Further, the relative importance of different sources or channels of communication about an innovation depends in part on what is available to the audience of potential adopters. For example, if a new idea is initially promoted only by the commercial firm that sells it, it is unlikely that other sources or channels will be very important, at least at the knowledge stage of the innovation-decision process. Coleman and others (1966) found that 80 percent of the medical doctors in their drug study reported first learning about gammanym from drug companies (57 percent from the pharmaceutical detail men, 18 percent from drug-house mailings, 4 percent from drug-house magazines, and 1 percent from drug ads in medical journals).

Later in the innovation-decision process, at the persuasion and decision stages, near-peer networks were the major sources or channels of communication about the new drug, and the commercial role was unimportant.* Awareness-knowledge that a new drug existed could be credibly communicated by commercial sources or channels, but doctors relied on the experiences of their peers for evaluative information about the innovation. The pharmaceutical firms that sold tetracycline were not regarded as credible by medical doctors.

*The Coleman and others (1966) finding that medical doctors did not perceive drug company detailmen as credible sources/channels of communication about a new drug was not supported by a replication study in Australia. Peay and Peay (1988) investigated the diffusion of temazepam, a new drug that was an important improvement in medical practice but not a "wonder drug" like tetracycline. The Australian doctors who had contact with detailmen (61 percent of the total sample) adopted the innovation earlier than did other doctors.

Scientific evaluations of tetracycline were communicated to the doctors, but such information did not convince them to adopt the innovation. Coleman and others (1966) concluded that, "The extensive trials and tests by manufacturer, medical schools, and teaching hospitals—tests that a new drug must pass before it is released—are not enough for the average doctor." They found that "teaching at the expert level cannot substitute for the doctor's own testing of the new drug; but testing through the everyday experiences of colleagues on the doctor's own level can substitute, at least in part." Individuals depend on their near-peers for innovation-evaluation information, which decreases their uncertainty about the innovation's expected consequences.

One type of evidence that the interpersonally communicated experience of near-peers can substitute, in part, for one's personal experience with an innovation is provided by analyses of the degree to which earlier versus later adopters of an innovation adopt a new idea completely at the time of their first trial. The first medical doctors to adopt gammanym did so on a very partial basis; the nineteen physicians who adopted the new drug in the first and second months of its use only wrote prescriptions for an average of 1.5 patients. The twenty-two doctors who adopted the innovation in the third or fourth months wrote 2.0 prescriptions, while the twenty-three doctors adopting in the fifth through eighth months wrote an average of 2.7 prescriptions (Coleman and others, 1966).*

Why are the first individuals in a system to adopt an innovation usually most tentative in their degree of trial of use of the new idea? The reason concerns the role of uncertainty in the diffusion process. Even though most innovative adopters of gammanym and hybrid corn were fully informed of the scientific evaluations that had been made of the new idea, this information did not reduce their uncertainty about how the innovation would work out for them. The innovators had to conduct their own personal experimentation with the new idea in order to assure themselves that it was indeed advantageous. They could not depend on the experience of peers with the innovation, because no one else had adopted the innovation at the time that

*Similarly, Ryan (1948) found that Iowa farmers adopting hybrid corn prior to 1939 initially planted only 15 percent of their corn acreage with hybrids, but those who adopted in 1939 and 1940 planted 60 percent of their acreage in hybrid seed in their first year of adoption. This figure was 90 percent for those starting in 1941–42 (the laggards).

the innovators adopted (at least in their system). Later adopters profit from their peers' accumulated personal experiences with the innovation; thus, much of the uncertainty of the innovation is removed by the time the later adopters first use a new idea, making a personal trial of the new idea less necessary for them.

This case illustration is based on Coleman and others (1966).

Categorizing Communication Channels

It is often difficult for individuals to distinguish between the source of a message and the channel that carries the message. A *source* is an individual or an institution that originates a message. A *channel* is the means by which a message gets from the source to the receiver. In the present section, we mainly speak of "channels" but often "source/channel" would be more accurate.

Researchers categorize communication channels as either (1) interpersonal or mass media in nature, or (2) originating from either local or cosmopolite sources. These channels play different roles in creating knowledge versus persuading individuals to change their attitude toward an innovation. The channels also are different for earlier adopters than for later adopters.

Mass media channels are means of transmitting messages involving a mass medium, such as radio, television, newspapers, and so on, that enable a source of one or a few individuals to reach an audience of many. Mass media can: (1) reach a large audience rapidly, (2) create knowledge and spread information, and (3) lead to changes in weakly held attitudes.

The formation and change of strongly held attitudes, however, is usually accomplished by interpersonal channels. *Interpersonal channels* involve a face-to-face exchange between two or more individuals. These channels have greater effectiveness in dealing with resistance or apathy on the part of the communicatee. What can interpersonal channels do best?

1. Provide a two-way exchange of information. One individual can secure clarification or additional information about the innovation from another individual. This characteristic of interpersonal networks often allows them to overcome the social-psychological barriers of selective exposure, perception, and retention.

2. Persuade an individual to form or to change a strongly held attitude. This role of interpersonal channels is especially important in persuading an individual to adopt an innovation.

Mass Media Versus Interpersonal Channels

Generalization 5–11 states: *Mass media channels are relatively more important at the knowledge stage and interpersonal channels are relatively more important at the persuasion stage in the innovation-decision process*. The importance of interpersonal and mass media channels in the innovation-decision process was first investigated in a series of researches with farmers, and then largely confirmed in studies of other types of respondents. For example, Sill (1958) found that if the probability of adoption were to be maximized, communication channels must be used in an ideal time sequence, progressing from mass media to interpersonal channels. Copp and others (1958, p. 70) found that "A temporal sequence is involved in agricultural communication in that messages are sent out through mass media directed to awareness, then to groups, and finally to individuals. A farmer upsetting this sequence in any way prejudices progress at some point in the adoption process." The greatest thrust out from the knowledge stage was provided by the use of the mass media, while interpersonal channels were salient in moving individuals out of the persuasion stage. Using a communication channel that was inappropriate to a given stage in the innovation-decision process (such as an interpersonal channel at the knowledge stage) was associated with later adoption of the new idea by an individual because such a channel use delayed progress through the process.

Data on the relative importance of interpersonal and mass media channels at each stage in the adoption of 2, 4-D weed spray were obtained by Beal and Rogers (1960) from 148 Iowa farmers. Mass media channels, such as farm magazines, bulletins, and container labels, were more important than interpersonal channels at the knowledge stage for this innovation. The percentage of respondents mentioning an interpersonal channel increased from 37 percent at the knowledge stage to 63 percent at the persuasion stage.

The evidence just presented in support of Generalization 5–11 came from research done in the United States, where the mass media are widely available. The mass media, however, may not be so widely available in Third World countries. For example, Deutschmann and Fals Borda (1962b) found that interpersonal channels were heavily used even at the knowledge stage by Colombian villagers. In Bangladesh villages, Rahim (1961, 1965) found that mass media channels were seldom mentioned as channels about agricultural innovations, whereas cosmopolite-interpersonal channels were very important, and in some ways seemed to perform a similar role to that played by mass media channels in more developed countries. An example of a cosmopolite-interpersonal channel is an Iowa farmer attending a farm machinery show in Des Moines, or a doctor traveling to an out-of-town medical specialty meeting.

Rogers with Shoemaker (1971) made a comparative analysis of the role played by mass media and cosmopolite-interpersonal channels by stages in the innovation-decision process for twenty-three different innovations (mostly agricultural) in the United States, Canada, India, Bangladesh, and Colombia. Mass media channels are of relatively greater importance at the knowledge stage in both developing and developed countries, although there was a higher *level* of mass media channel usage in the developed nations, as we would expect. Mass media channels were used by 52 percent of the respondents in developed nations at the persuasion stage, and 18 percent at the decision stage. The comparable figures for respondents in Third World nations were 29 percent and 6 percent. This meta-research showed that cosmopolite-interpersonal channels were especially important at the knowledge stage in developing nations.

Cosmopolite Versus Localite Channels

Generalization 5–12 states: *Cosmopolite channels are relatively more important at the knowledge stage, and localite channels are relatively more important at the persuasion stage in the innovation-decision process.* Cosmopolite communication channels are those from outside the social system of study; interpersonal channels may be either local or cosmopolite, while mass media channels are almost entirely cosmopolite. The meta-research for twenty-three different innovations in ten nations (mentioned previously) showed that if cosmopolite-interpersonal and mass media channels are combined to form the category of cosmopolite channels, in the Third World nations the percentage of such channels was 81 percent at the knowledge stage and 58 percent at the persuasion stage. In developing nations, the percentages were 74 percent at the knowledge stage and 34 percent at the persuasion stage. These meta-research data hint that the role played by mass media channels in developed countries (creating awareness-knowledge) is perhaps partly replaced by cosmopolite-interpersonal channels in developing countries. These channels include change agents, visits outside the local community, and visitors to the local system from the city.

Communication Channels by Adopter Categories

The preceding discussion of communication channels by stages in the innovation-decision process ignored the effects of the respondent's adopter category. Now we explore channel usage by different adopter categories.

Generalization 5–13 states: *Mass media channels are relatively more important than interpersonal channels for earlier adopters than for later adopters.* At the time that innovators adopt a new idea there is almost no one else in the system who has experience with the innovation. Later adopters do not need to rely so much on mass media channels because a store of interpersonal, local experience has accumulated in their system by the time they decide to adopt. Perhaps interpersonal influence is not so necessary to motivate earlier adopters to decide favorably on an innovation. They possess a more venturesome orientation, and the mass media message stimulus is enough to move them over the mental threshold to adoption. But the less change-oriented later adopters require a stronger and more immediate influence, like that from interpersonal networks.

There is strong support for Generalization 5–13 from researches in both developed and developing nations. Data illustrating this proposition are shown in Figure 5–2 for the adoption of a chemical weed spray by Iowa farmers.

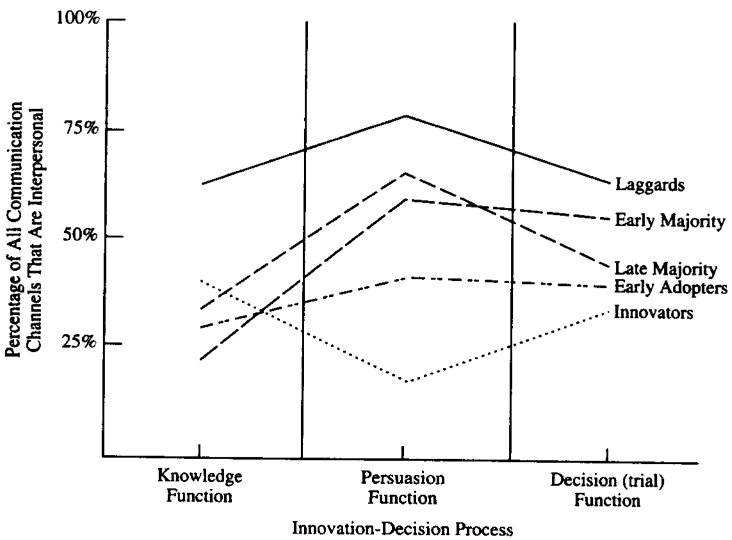
Reasoning similar to that just presented leads to Generalization 5–14: *Cosmopolite channels are relatively more important than localite channels for earlier adopters than for later adopters.** Innovations enter a system from external sources; those who adopt first are more likely to depend upon cosmopolite channels. These earlier adopters, in turn, act as interpersonal and localite channels for their later-adopting peers.

The Innovation-Decision Period

The *innovation-decision period* is the length of time required for an individual or organization to pass through the innovation-decision

*This proposition bears close resemblance to Generalization 7–25, which states that earlier adopters are more cosmopolite than later adopters. Generalization 5–14, however, refers to cosmopolite *channel* usage, rather than to cosmopolite behavior in general.

Figure 5–2. Interpersonal Channels Are Relatively Less Important for Earlier Adopters Than for Later Adopters of 2, 4-D Weed Spray in Iowa



Source: Beal and Rogers (1960).

process.* The time elapsing from awareness-knowledge of an innovation decision for an individual is measured in days, months, or years. The period is thus a gestation period during which a new idea ferments in an individual's mind.

Rate of Awareness-Knowledge and Rate of Adoption

Most change agents wish to speed up the process by which innovations are adopted. One method of doing so is to communicate information about new ideas more rapidly or more adequately so that knowledge is created at an earlier date. Another method is to shorten the amount of

*The length of the innovation-decision period is usually measured from first knowledge until the decision to adopt (or reject), although in a strict sense it should perhaps be measured to the time of confirmation. This later procedure is often impractical or impossible because the confirmation function may continue over an indefinite period.

time required for the innovation-decision after an individual is aware of a new idea. Many potential adopters are aware of an innovation but are not motivated to try it. For example, almost all of the Iowa farmers in the hybrid corn study heard about the innovation before more than a handful were planting it. "It is evident that . . . isolation from knowledge was not a determining factor in late adoption for many operators" (Ryan and Gross, 1950). Shortening the innovation-decision period is thus one of the main methods of speeding the diffusion of an innovation.

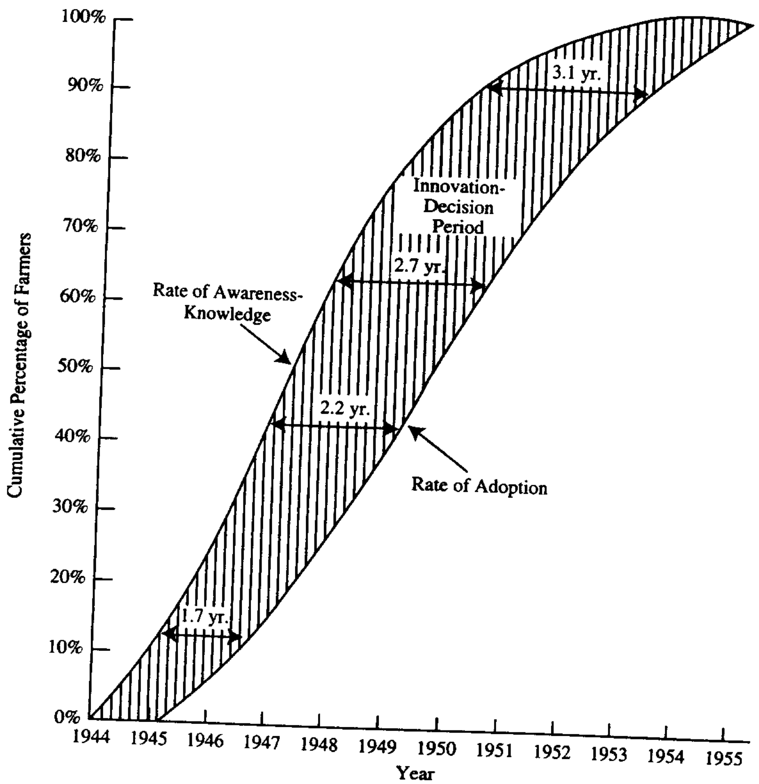
Figure 5–3 illustrates the interrelationships between the rate of awareness-knowledge, rate of adoption, and the innovation-decision period for a new weed spray. The slope of the curve for the rate of awareness is steeper than that for the rate of adoption. These data, along with evidence from other studies, suggest Generalization 5–15: *The rate of awareness-knowledge for an innovation is more rapid than its rate of adoption.* Later adopters have longer innovation-decision periods than earlier adopters, a point to which we shall soon return.

There is a great deal of variation in the average length of the innovation-decision period from innovation to innovation. For instance, 9.0 years was the average period for hybrid corn in Iowa (Gross, 1942), while 2.1 years was the average for weed spray depicted in Figure 5–3 (Beal and Rogers, 1960). How can we explain these differences? Innovations with certain characteristics are generally adopted more quickly; they have a shorter innovation-decision period. For example, innovations that are relatively simple in nature, divisible for trial, and compatible with previous experience usually have a shorter innovation-decision period.

Length of the Period by Adopter Category

One of the important individual differences in length of the innovation-decision period is on the basis of adopter category. We pointed out previously that the data in Figure 5–3 show a longer period for later adopters. We show this relationship in greater detail in Figure 5–4, where the average length of the period is shown for the five adopter categories. These data and those from other studies support Generalization 5–16: *Earlier adopters have a shorter innovation-decision period than later adopters.* Thus the first individuals to adopt a new idea (the innovators) do so not only because they become aware of the innovation somewhat sooner than their peers, but also because they require fewer months or years to move from knowledge to decision. Innovators perhaps gain part of their innovative position (relative to late adopters) by learning about innovations at

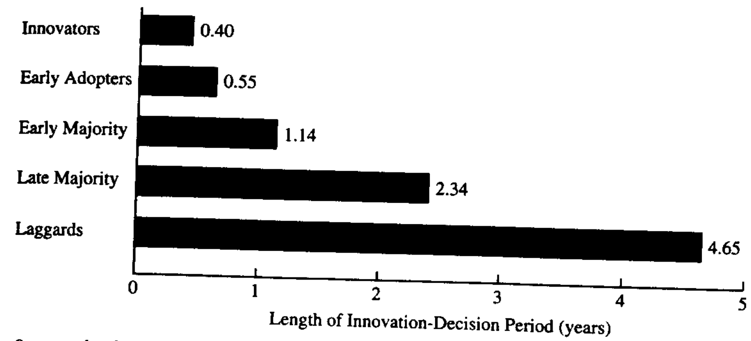
Figure 5–3. Rate of Awareness-Knowledge, Rate of Adoption, and Length of the Innovation-Decision Period for Iowa Farmers Adopting a Weed Spray by Year



The shaded area in this figure illustrates the aggregate innovation-decision period between awareness-knowledge and adoption of weed spray. Knowledge proceeds at a more rapid rate than does adoption, a finding that suggests that relatively later adopters have a longer average innovation-decision period than earlier adopters. For example, there are 1.7 years between 10 percent awareness and 10 percent adoption, but 3.1 years between 92-percent awareness and 92 percent adoption.

Source: A reanalysis of data originally gathered by Beal and Rogers (1960).

Figure 5–4. Innovators Have Shorter Innovation-Decision Periods Than Laggards in Adopting 2, 4-D Weed Spray



Source: This figure is based on data from 148 Iowa farmers, gathered by Beal and Rogers (1960).

an earlier time, but the present data suggest that innovators are the first to adopt because they require a shorter innovation-decision period.

Why do innovators require a shorter period? Research studies show that innovators have more favorable attitudes toward new ideas and so less resistance to change must be overcome by communication messages about innovations. Innovators may also have shorter innovation-decision periods because (1) they use more technically accurate sources and channels about innovations, such as direct contact with scientists, and (2) they place higher credibility in these sources than the average individual. Innovators may also possess a type of mental ability that better enables them to cope with uncertainty and to deal with abstractions. An innovator must be able to conceptualize relatively abstract information about innovations and apply this new information to his or her own situation. Later adopters can observe the results of innovations by earlier adopters and may not require this type of mental ability.

Summary

The *innovation-decision process* is the process through which an individual (or other decision-making unit) passes from first knowledge of an innovation, to forming an attitude toward the innovation, to a decision to adopt or reject, to implementation of the new idea, and to confirmation of this decision. This process consists of five stages: (1) *knowledge*—the individual (or decision-making unit) is exposed to the innovation's existence and gains some understanding of how it functions; (2) *persuasion*—the individual (or other decision-making unit) forms a favorable or unfavorable attitude toward the innovation; (3) *decision*—the individual (or other decision-making unit) engages in activities that lead to a choice to adopt or reject the innovation; (4) *implementation*—the individual (or other decision-making unit) puts an innovation into use; and (5) *confirmation*—the individual (or other decision-making unit) seeks reinforcement for an innovation-decision already made, but may reverse this decision if exposed to conflicting messages about the innovation.

Earlier knowers of an innovation, when compared to later knowers, are characterized by more formal education, higher social status, greater exposure to mass media channels of communication, greater exposure to interpersonal channels of communication, greater change agent contact, greater social participation, and more cosmopoliteness (Generalizations 5–1 to 5–7).

Re-invention is the degree to which an innovation is changed or modified by a user in the process of its adoption and implementation. Re-invention occurs at the implementation stage for certain innovations and for certain adopters (Generalization 5–8).

Discontinuance is a decision to reject an innovation after having previously adopted it. Two types of discontinuance are: (1) *replacement discontinuance*, in which an idea is rejected in order to adopt a better idea that superseded it, and (2) *disenchantment discontinuance*, in which an idea is rejected as a result of dissatisfaction with its performance. Later adopters are more likely to discontinue innovations than are earlier adopters (Generalization 5–9).

We conclude, on the basis of research evidence, that stages exist in the innovation-decision process (Generalization 5–10). Needed in the future is *process research*, a type of data gathering and analysis that seeks to determine the time-ordered sequence of a set of events. Most past diffusion study has been *variance research*, a type of data gathering and analysis that consists of determining the covariances among a set of variables but not their time-order.

A *communication channel* is the means by which a message gets from a source to a receiver. We categorize communication channels (1) as either interpersonal or mass media in nature, and (2) as originating from

either local or cosmopolite sources. *Mass media channels* are means of transmitting messages involving a mass medium such as radio, television, newspapers, and so on, that enable a source of one or a few individuals to reach an audience of many. *Interpersonal channels* involve a face-to-face exchange between two or more individuals.

Mass media channels are relatively more important at the knowledge stage, and interpersonal channels are relatively more important at the persuasion stage in the innovation-decision process (Generalization 5–11). Cosmopolite channels are relatively more important at the knowledge stage, and local channels are relatively more important at the persuasion stage in the innovation-decision process (Generalization 5–12). Mass media channels are relatively more important than interpersonal channels for later adopters (Generalization 5–13). Cosmopolite channels are relatively more important than local channels for earlier adopters than for later adopters (Generalization 5–14).

The *innovation-decision period* is the length of time required for an individual or organization to pass through the innovation-decision process. The rate of awareness-knowledge for an innovation is more rapid than its rate of adoption (Generalization 5–15). Earlier adopters have a shorter innovation-decision period than do later adopters (Generalization 5–16).